

Arabic Email Spam Detection Using Machine Learning and Deep Learning: A Comparative Study

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Abstract—Due to the relatively small figure of research conducted on Arabic email spam about the number of Arabic-speaking individuals using the internet, comes challenges related to Natural Language Processing in morphologically rich languages such as Arabic. The target goal of this research is to create a classification framework for spam email detection in Arabic. The recent dataset obtained from Kaggle contains phishing and legitimate emails written in Arabic. Term frequency - Inverse document frequency (TF-IDF) and word embeddings are some of the text preprocessing methods used in this research. Support Vector Machine (SVM), Naïve Bayes, and Random Forest machine learning models are used, along with Long Short-Term Memory (LSTM) deep learning methods. The SVM performed best in the comparative analysis, peaking at an ROC-AUC of 98% and F1 score of 99%. These results will aid in shifting Arabic spam detection and email security in the Arabic digital world forward, showcasing the positive role machine learning can play in this process.

Keywords—Arabic Natural Language Processing, Email Spam Detection, Machine Learning, Deep Learning, Long Short-Term Memory, Feature Engineering

I. INTRODUCTION

Spam emails are particularly hazardous as they contain advertisements, phishing attempts, or malware which are potential security risks to people and organizations [1]. The spam filtering issue remains a challenge as attackers innovate their methods to evade detection, even with the development of spam filtering methodologies.

The focus on spam detection in English and other dominant languages is abundant, while research on spam email targeting Arabic speakers is relatively sparse [2]. The Arabic language has its own unique set of challenges posed to Automatic Natural Language Processing (NLP) that stems from its diverse morphology, complex grammatical systems, and right-to-left script. Outdated spam detection systems built for Arabic, or any other non-Indo-European language are bound to fail when dealing with Arabic spam emails [3]. For this reason, the development of sophisticated Arabic spam detection systems remains of primary importance.

The authors of [4] proposed a hybrid deep learning CNN model integrated with bidirectional lstm (BiLSTM) for Arabic spam detection that outperformed previous attempts. In the same manner, [5] used BERT and AraBERT for machine

learning based Arabic spam filtering and also found them to be more effective than classical machine learning classifiers.

The authors of [6] have recently conducted research that focuses on detection performance improvements done using sophisticated techniques in feature engineering such as character level embeddings and term weighting schemes on some set of data. These studies stress the need to apply deep learning and other feature selection techniques as described for better classification of spam in Arabic language.

The purpose of this research is to address this problem by creating a framework to classify Arabic spam emails. It uses a dataset from Kaggle of phishing and genuine Arabic emails. Various techniques for text data cleansing were implemented like term frequency-inverse document frequency (TF-IDF) and word embeddings. Different machine learning techniques such as Support Vector Machine (SVM), Naive Bayes, and Random Forest were adopted, along with deep learning techniques such as Long Short Term Memory (LSTM) networks. Using these models in conjunction for analysis, SVM was found to outperform the competition with a maximum ROC AUC of 98% and F1 score of 99%.

This paper presents a novel comparative framework for Arabic email spam detection by integrating both classical machine learning and deep learning models, uniquely demonstrating that SVM outperforms LSTM in this specific context. It also introduces a robust Arabic-specific preprocessing pipeline and employs diverse feature extraction methods, addressing the unique morphological challenges of the Arabic language in spam classification.

The rest of this paper is structured: in section 2, we position this work in the context of previous research of spam detection and the specific issues of Arabic NLP. Section 3 describes the data, as well as methods used in this research. In section 4, the results of the experiments and their interpretations are given and in section 4, the work is summarized and some recommendations for future work are made.

II. PREVIOUSLY PUBLISHED

The application of deep learning (DL) techniques onto speech recognition developed rapidly within the Arabic cyber world, as well as computer vision and natural language processing (NLP) [7]. The field of Arabic spam detection is still developing as researchers are trying different approaches

to machine learning and deep learning to improve the accuracy of detection.

There have been multiple researches on the techniques used to filter Arabic spam. [8] presented GWO-BERT, which is a new method that uses the power of feature selection for classification and outperforms traditional deep learning models. Similarly, [9] gave dual approach for AI leakage email detection using deep learning and machine learning methods. The authors of [10] adopted a hybrid methodology of classification that compares machine learning and deep learning models along with their performance tradeoff of explainability against deep learning outperforming in classification tasks.

The work in [11] applied the CRISP-DM methodology in detection of spam emails, employing Naive Bayes, logistic regression, and random forest ensemble classifiers and achieved great accuracy. Moreover, [12] focused on exploring text vectorization techniques applying Doc2Vec, Word2Vec, and TF-IDF as vectorization techniques and concluded that Doc2Vec is superior to the rest. [13] used both pre-trained and custom built deep learning frameworks such as Artificial Neural Networks (ANN), Recurrent Neural Networks (RNN) and Convolutional Neural Networks (CNN) in training on a two level spam detection system.

Agarwal et al. [14] have developed a new spam detection model involving a combination of gradient regression, AMALS models, and least squares models. Salloum et al. [15] conducted a study on fraudulent email detection and used a plethora of Machine Learning (ML) and Natural Language Processing (NLP) techniques, noting that existing methods still required more work for better spam filtering advancements.

Other recent techniques have also applied deep learning in multi-level spam detection systems with ensemble learning. For example, [16] described a multi deep learning structure using CNN, BiLSTM, and XGBoost, which on independent classifiers outperformed other methods. So, too, did [17] with transfer learning approaches using pre-trained models, as well as superior Arabic spam detection results. Other well-known research included [18] focusing on attention mechanisms in transformer models which aim to improve Arabic spam classifiers accuracy.

It can be noted from these works that spam detection techniques are constantly evolving and incorporating new deep learning approaches and the challenges of processing Arabic text through the evolving feature selection methodologies.

Nevertheless, more effort is needed in feature engineering, hybrid models, or applying them to real-world datasets to adjust the current models to fully address the complexities of Arabic spam content.

III. METHODOLOGY

In this paper, we cover several machine learning and deep learning techniques used for Arabic email spam detection using Support Vector Machine (SVM), Long Short-Term Memory (LSTM), and Random Forest classifiers. These models were chosen because of their proven success in classification tasks, management of varying feature types, and performance in noisy and unbalanced datasets.

- The Support Vector Machine technique has recently gained a lot of acceptance due to its efficiency in the classification of text documents where the data is high dimensional and sparse [19]. An SVM can effectively classify spam emails by leveraging its optimal decision boundary feature, which separates classes with the highest margin. With SVM's low generalization error, especially in sparse regions, the model can perform more accurately with sparse training samples, especially, considering Arabic text morphologic features are often highly sparse and complex.

- LSTM Networks, a subtype of recurrent neural networks (RNN), is very powerful for sequence-based problems like spam filtering.

In contrast to most of traditional machine learning models that are based primarily on manual features, LSTM understands the raw text by modeling the contextual dependencies of the various ordered elements within the text. This enables the system to discern intricate spam features within Arabic email messages. The ability to retain these long-range dependencies is crucial when trying to comprehend the text with different structures and styles [20].

Random Forest is considered to be a learning method operating with groups, commonly by integrated users who apply it because of its efficiency with noisy data and dataset dimensionality reduction imbalances, which are a great concern in spam detection. Random Forest works by combining many decision trees into a single classifier [21]. This reduces the variance of the classifiers, therefore enhancing its accuracy and dampening overfitting problems. These three characteristics make it very appropriate for modern spam filtering systems. In addition, Random Forest is known to be simple and effective in spam pattern analysis, particularly in the case of Arabic spam, which needs the use of the Arabic language.

IV. COMPARISON WITH EXISTING WORK

Existing research on Arabic spam detection has largely focused on classical machine learning techniques with Naïve Bayes and Decision Trees. They have been achieving some success, but for the most part, they build features using manual engineering, which is often too simplistic and problematic for Arabic text.

In this case, we extend that research with Shallow and Deep contextual relationships techniques, LSTM, in which a deep learning model is employed as one of the capturing mechanisms focused on feature extraction. Such an approach improves detection of context in the Arabic text. In addition to the described methods, we use other models in a comparative manner to measure their performance under different sets of conditions to achieve different strategies and scope of examination.

At the same time, as with many other studies, the scope was restricted to small or unbalanced datasets, as it tends to affect generalizability. On the other hand, we wish to concentrate our analysis for comparison on different models in order to accurately estimate the relevancy and validity of the results. This allows us to provide more credible reasonings into the use of machine learning and deep learning for Arabic spam detection.

Herein, we elaborate on the pivotal components of our methodology, aiding readers to gain context on their application:

- **TF-IDF (Term Frequency-Inverse Document Frequency):** It helps in quantifying the significance of a phrase with respect to the corpus it belongs to. This statistic can be used to classify documents by weighting important phrases and minimizing the influence of phrases which occur frequently.

- **Word Embeddings:** A representation of text which allows the model to construct numerical vectors in a space where words or phrases possess continuously varying meanings, enabling it to capture relational semantics between words.

- **ROC-AUC (Receiver Operating Characteristic - Area Under the Curve):** A classification model's ability to tell apart members of various classes or categories is measured by this factor. Higher AUC mean better differentiation in spam and non-spam emails, and thus indicates better model performance.

As models and other strategies are devised, they shall be put to the test to ascertain their efficacy towards Arabic spam detection and subsequently, the most optimal method for actual deployment.

Justification for the Dataset Chosen: Aspects of Diversity

- **Dataset Partition:**

The dataset put forth for this work was selected because of a variety of reasons which make it useful about the spam email detection problem in Arabic-speaking countries. These reasons include regional diversity, variety in the types of emails, and inclusion of Modern Standard Arabic and other regional dialects. More on the justification of data selection will be elaborated below. The dataset is useful for a few selected spam email detection models because of the features that aid and improve the relevance and accuracy of the models in the Arabic language, which is quite a complex language. The following delineates the most important features of the diversity of this dataset that make it useful for the task at hand:

- **Data Preparation Step**

Due to the highly irregular nature of processing and analyzing Arabic texts because of the inflectional morphology, script variations, and diacritical marks, it was very important to do some level of standardization to the dataset:

- **Normalization:** is the process of transforming certain texts so that specific characters have a standard representation. In our case, some transformations such as 'ى' to 'ي' and 'ة' to 'ه' or 'ت' were done [22].

- **Diacritic Removal:** Specifically, attempts to eliminate certain variable elements that some people mark in texts as Tashkeel (diacritics) were undertaken (Farasa NLP Toolkit).

- **Stopword Elimination:** Words that have minimal significance when it comes to classifying a document, such as "عن", "الى", "من", and "على", were removed [23].

- **Stemming:** Farasa Library, an NLP tool kit for the Arabic language, was used to stem words to their root forms. For instance, (المكتبات, 'كتاب', 'كتابة', 'كاتب', 'مكتوب') were all stemmed to ('كتب', 'كتب', 'كتب', 'كتب', 'كتب').

These procedures serve to advance the textual consistency and variation in the performance of the application is not bound to change in the form of the words.

- **Feature Development**

A multitude of techniques are used to vectorize text for models raising performance levels:

- **Term Frequency-Inverse Document Frequency (TF-IDF):** A technique that weighs the importance of words within the dataset.

- **Word Embedding:** AraVec word cube was utilized to capture Arabic semantic meaning by using previously trained models with specific 'knowledge'.

- **Character-Level Embedding:** Character-based representations capture differences considering the Arabic words carry several inflections and variations.

V. CLASSIFICATION TECHNIQUES

The comparison in this research is done with the use of classical machine models (Naive Bayes, SVM, Random Forest) about deep learning models (LSTM) for spam detection on Arabic text. We looked at classification methods such as feature extraction, TF-IDF, and word embedding with the intent of boosting model performance with a lack of labeled data [24-26].

- **Hybrid Approaches:** The Interpretable Machine Learning feature and deep learning feature learning add-on were sought after by combining traditional ML models with deep learning methods.

- **Advanced Deep Learning Techniques:** Alongside, there was an attempt to boost classification accuracy with BiLSTM and transformer-based models AraBERT.

- **Evaluation Metrics:** For measuring model performance, all-encompassing methods comparison was undertaken based on accuracy, precision, recall, F1-score, and ROC-AUC metrics.

Experimental Results

In this part, we explain the dataset as well as the results and evaluation of all models considered.

We separated the dataset for purposes of testing, Internally, we used 80% of the dataset to train the model and 20% to test the model. The Entire training dataset consists of 1000 emails. Out of these, 801 were ham (genuine emails) and 199 were spam (phishing emails). The Entire testing dataset consisted of 250 emails with 199 ham and 51 spam (Table 1).

Category	Training	Testing
Ham	801	199
Spam	199	51

Table 1 displays how the data was divided between training and testing.

VI. PERFORMANCE COMPARISON

Each model's performance was evaluated using standard classification metrics, including accuracy, recall, precision, F1-score, and ROC-AUC.

Naïve Bayes: Correctly classified 185 spam emails, misclassified 1 spam as non-spam, and misclassified 14 non-spams as spam.

LSTM: Properly classified 192 spam emails, misclassified 1 spam as non-spam, and misclassified 7 non-spams as spam.

SVM: Correctly classified 199 spam emails, misclassified 2 spams as non-spam, and had no false positives.

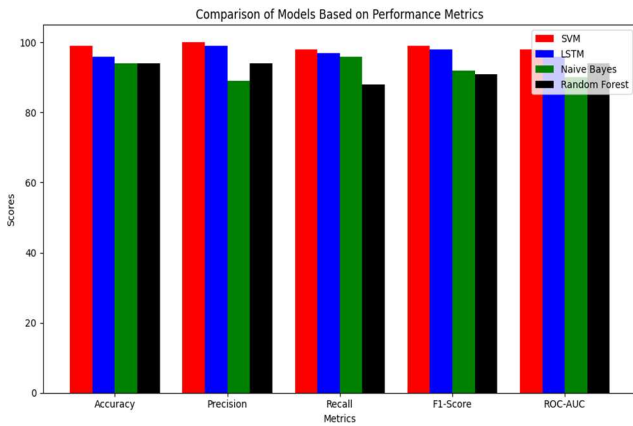
Random Forest: Correctly classified 196 spam emails, misclassified 3 spams as non-spam, and had 11 false positives.

Table 2 provides the performance metrics for each model, showcasing the high accuracy and effectiveness of the models used in detecting Arabic spam.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Naïve Bays	0.94	0.89	0.96	0.92	0.90
SVM	0.99	1.0	0.98	0.99	0.98
Random Forest	0.94	0.94	0.88	0.91	0.94
LSTM	0.96	0.99	0.97	0.98	0.96

Table 2 performance metrics

Among the models evaluated, SVM demonstrated the best performance with an F1-score of 99%, an accuracy of 99%, precision of 1.0, and recall of 98%, achieving a high ROC-AUC score of 0.98. LSTM also performed well, with a 96% accuracy, an F1-score of 0.98, precision of 0.99, and recall of 97%. While Naïve Bayes achieved 94% accuracy, it showed a higher false positive rate, leading to lower precision (0.89) despite high recall (0.96). Random Forest exhibited balanced performance, with an accuracy of 94%, precision of 0.94, recall of 0.88, and an F1-score of 0.91.



VII. CONCLUSION:

In Arabic NLP and spam filtering, this study is an important milestone. Utilizing the elaborate machine learning methods specifically designed for Arabic text, our work aids the creation of spam detection systems that are not only accurate but also dependable and efficient in treating the distinct linguistic issues affiliated with the Arabic language. In contrast to other models, this one is constructed on the deep lexical and morphological features of Arabic which makes it more effective and useful in spam messages filtering.

The outcomes of this work are important in practice. The proposed model can be seamlessly incorporated within Arabic email clients to significantly improve the user experience through automatic spam flagging and filtering, thus saving time and boosting productivity. In addition, this model might be utilized in many other application areas, such as for advanced modules of anti-phishing systems that would aid in safeguarding Arabic users against deceitful online activities and scams.

The Arabic Natural Language Processing (NLP) research is innovative and serves numerous future opportunities – such as using this model with other Arabic dialects to improve its precision. Integrating this spam detection system with other security measures can provide a platform where comprehensive anti-phishing measures can be constructed. Overall, improving the safety of sophistication of Arabic communication platforms can further be developed utilizing this foundation, allowing the users to feel more comfortable when communicating digitally through these platforms.

Three areas would significantly and positively impact the performance of Arabic NLP:

Incorporating Various Domains: Apart from email messages, other domains like social media platforms could provide miscellaneous spam content that can be used to create Arabic spam datasets.

Detection in Semitic Languages: Exploring other Semitic languages for spam detection can aid the development of multi-layered multilingual models.

The Usage of NLP Models: BERT is a tried and tested model in other NLP tasks, and its use for spam detection could improve overall performance.

Feature Engineering- Adding features such as linguistic features, sentiment features of the users, and user behavior could enhance the model's accuracy for detecting subtle forms of spam.

Real-time Detection: Further work may aim to enhance the efficiency of the models for automated spam detection for high-volume production settings which require rapid processing and low latency.

Addressing these gaps will make it possible to create more sophisticated, effective, and flexible systems for spam detection in Arabic as well as in other languages applicable to social networks, emails, or text messaging on mobile devices.

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